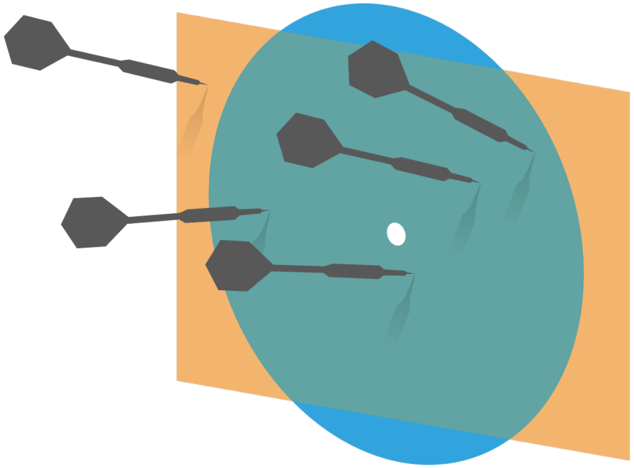
# Probability and Statistics for Computer Science



Credit: wikipedia

“The statement that “The average US family has 2.6 children” invites mockery” – Prof. Forsyth reminds us about critical thinking

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# PollEv site

PollEv.com/hongyeliu713

# Last lecture

- Welcome/Orientation
- Big picture of the contents
- Lecture 1 - Data Visualization & Summary (I)

# Q. Which of the following data is not categorical?

A. Number of enrolled students in a class
B. Weight of apples in a grocery store
C. Instruments played by an orchestra
D. Type of chemical reagents in a lab
E. A & B

# Objectives

- **Simple visualization** of 1-D data
- **Grasp** Summary Statistics
- Learn more Data Visualization for **Relationships**

# Simple Visualization of Data

- General principles
- Bar chart
- Histogram
- Conditional histogram

# Simple Visualization of Data

- General principles

  Must not mislead or distort;

  Aesthetically pleasing;

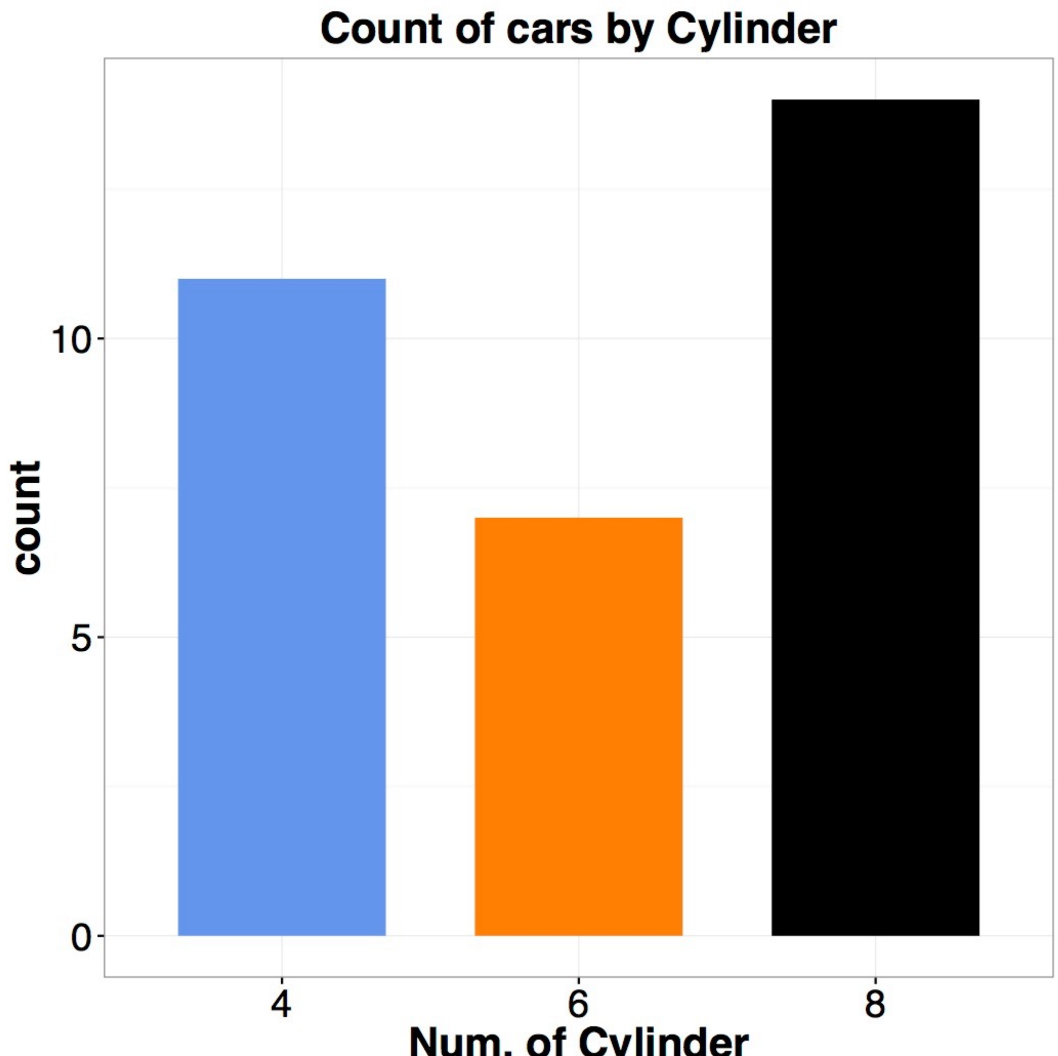
  Clear, Attractive, Convincing;

  Show message/significance.

# Simple Visualization of Data

- Bar chart

*A set of bars that are organized by categorical or ordinal feature*




Data: "mtcars"

# An example of good, ugly, bad, wrong

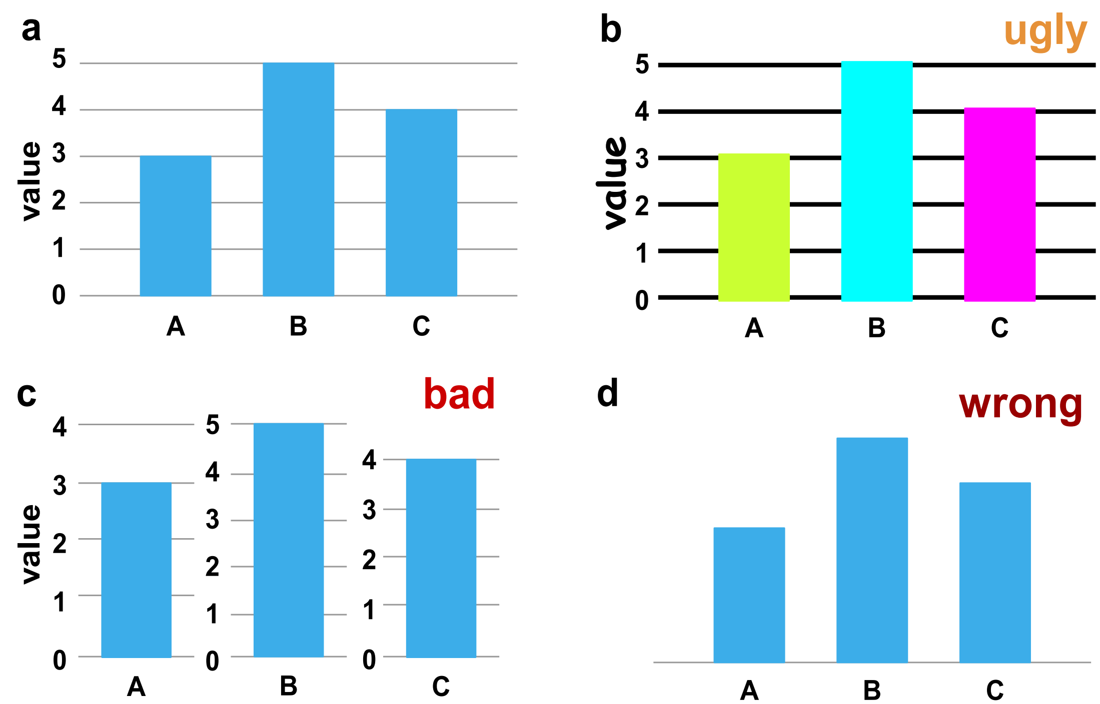
The difference between ***good*, *ugly*, *bad* and *wrong* visualization**

*a – good*

*b – ugly*

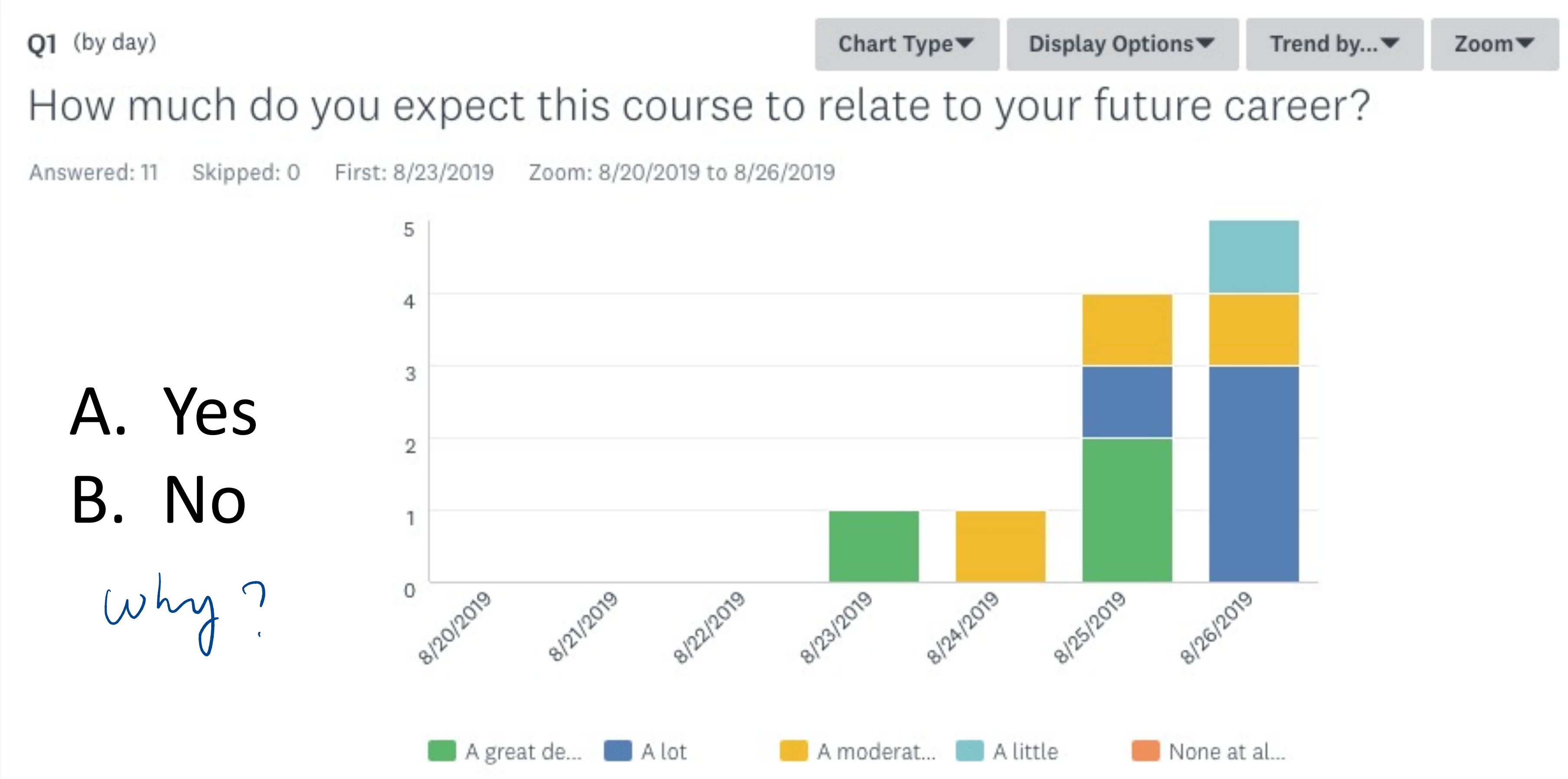
*c – bad*

*d - wrong*



Do this at home

# Q: Is this a good bar chart? why?

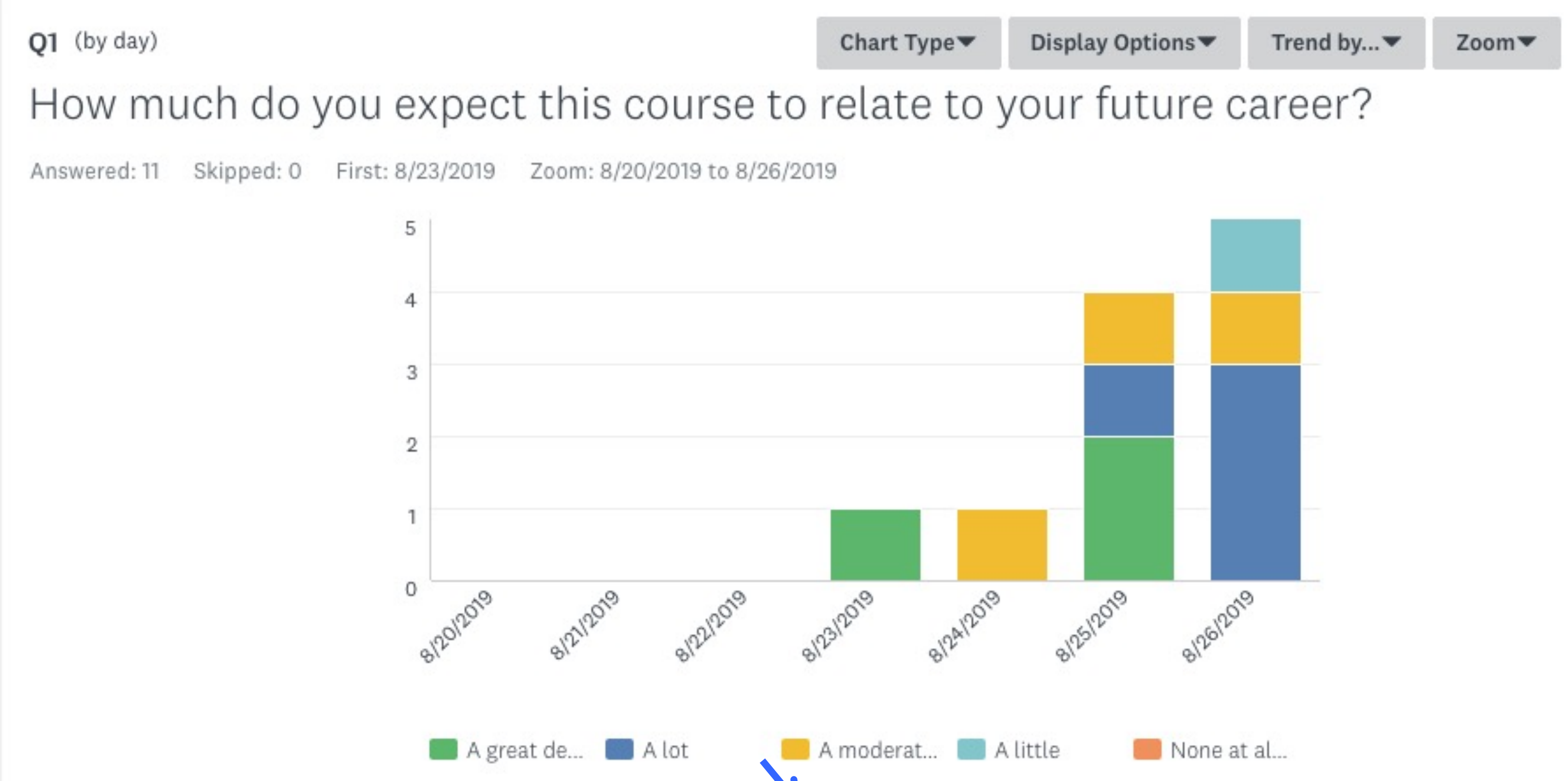



A. Yes
B. No

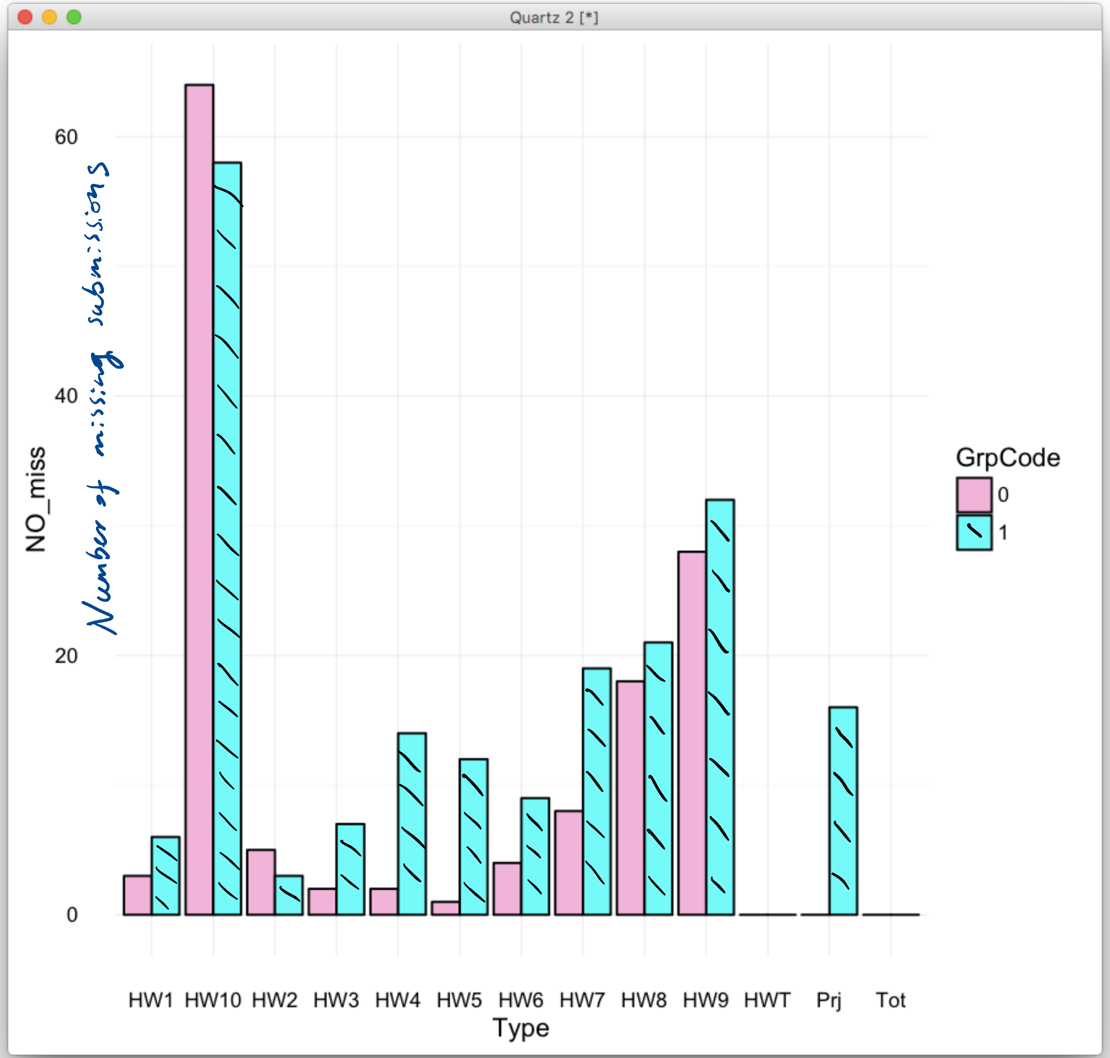
why?

Review at home

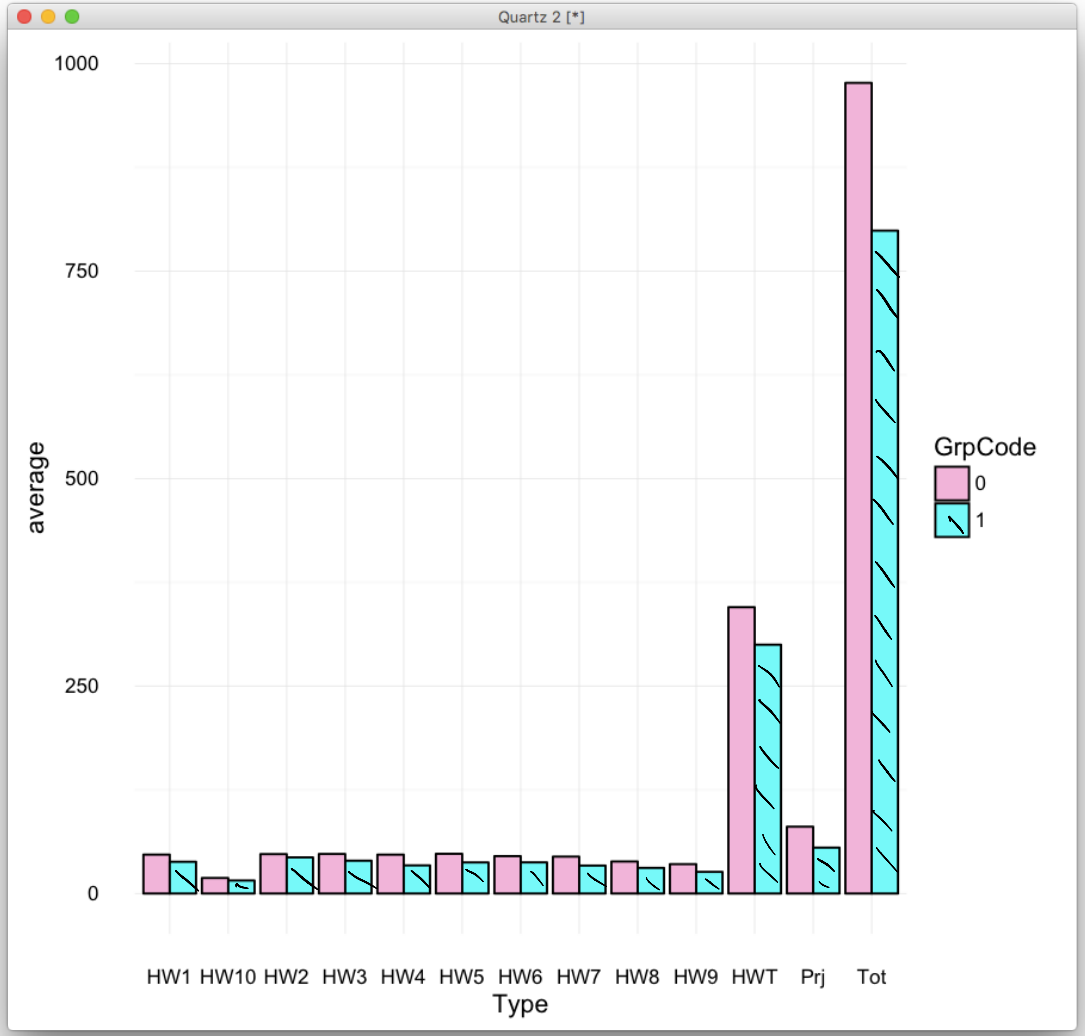
# How about using a color scale




# Which group has the higher total scores?



# Which group has the higher total scores?

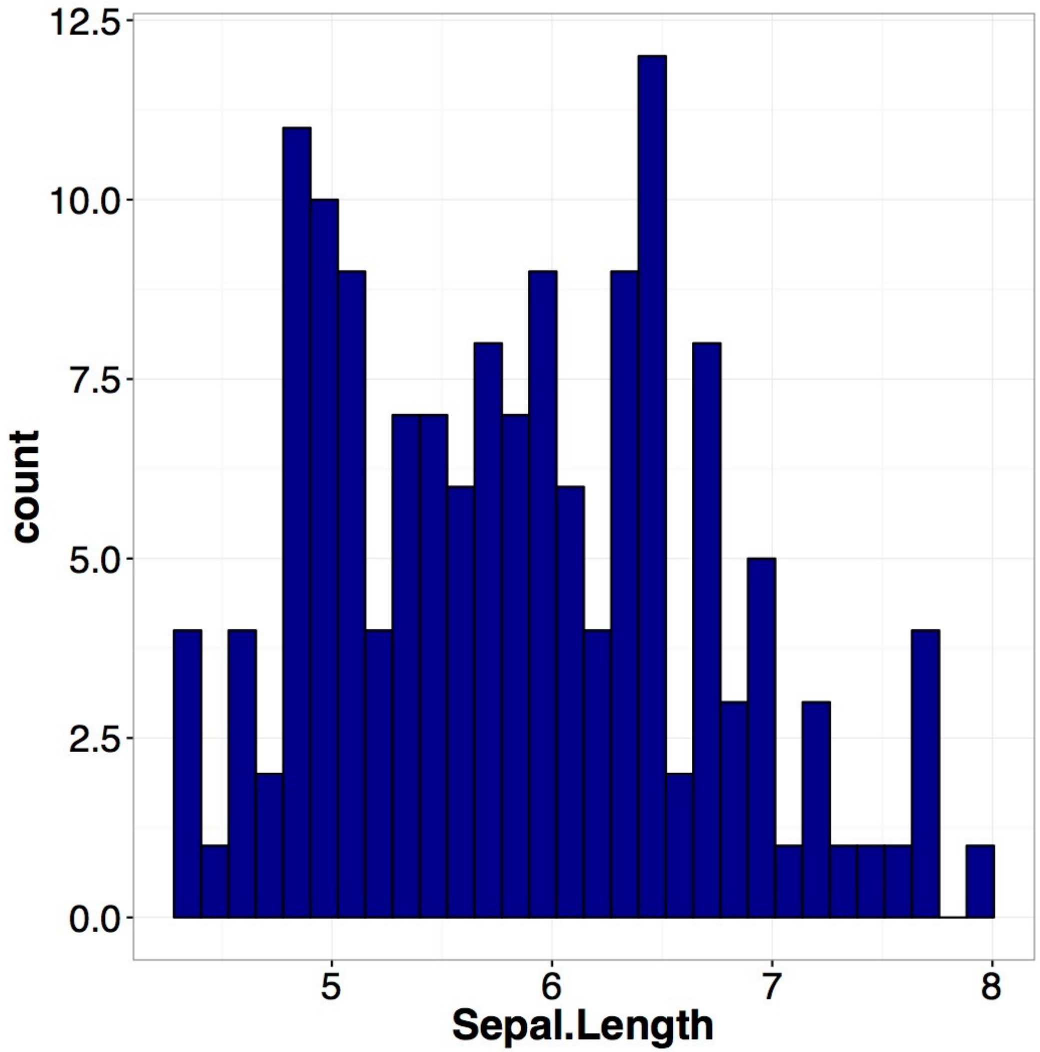


# Visualizing Data with Histogram

- Histogram

*A set of bars that are organized by bins that contains numerical data (discrete or continuous)*

Data: “iris”

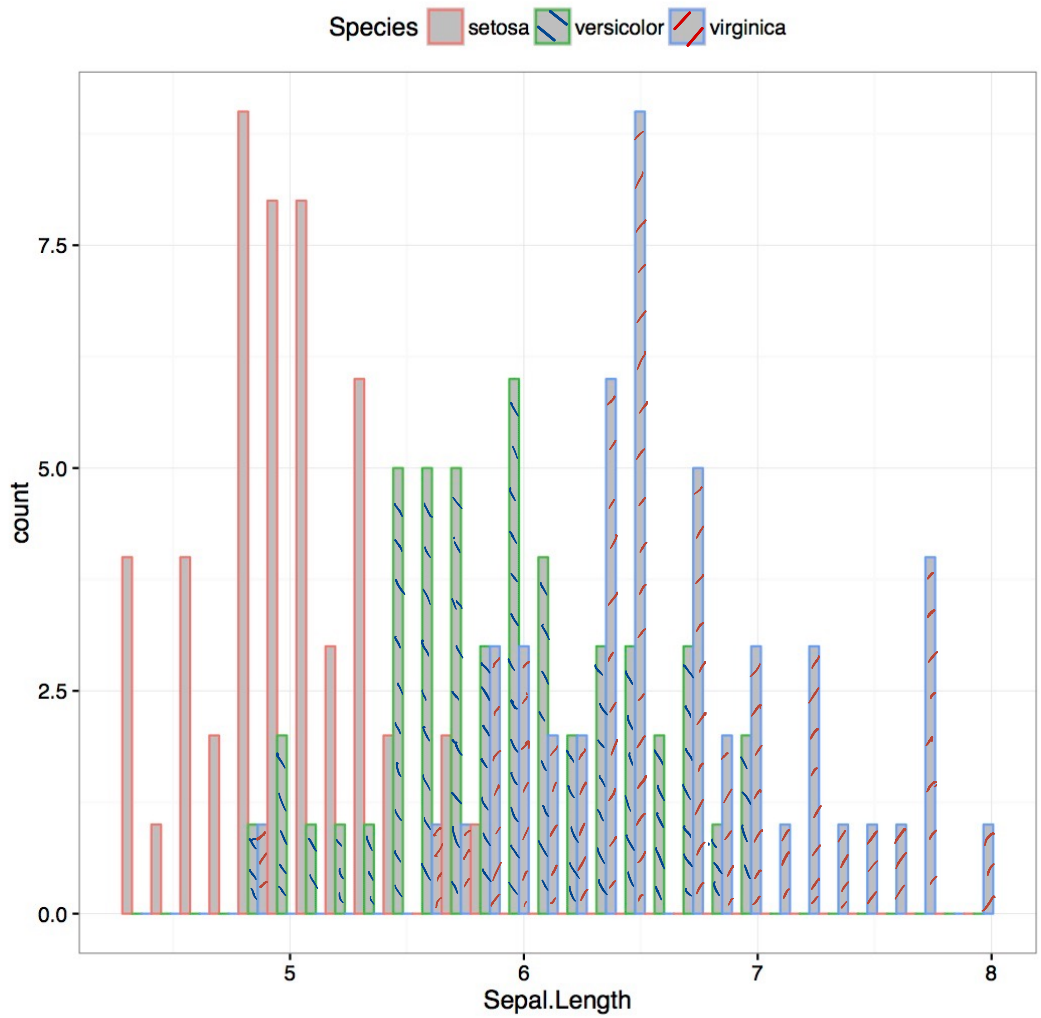


# Visualizing Data with Histogram (II)

- Conditional histogram

*Histogram generated by subsets of the data*

Data: “iris”



# Summarizing 1D continuous data

For a data set {x} or annotated as $\{x_i\}$, we summarize with: $\{x\} = \{x_i\}_{i \in 1:N}$

- Location Parameters 3 M

  Mean Median Mode

- Scale parameters

  Std Variance IQR

# Descriptive stats of 1-D continuous data

Location

Scale

| Data set | Mean | Median | Mode | Standard deviation | Variance | Interquar-tile range |
|---|---|---|---|---|---|---|
| {x} or $\{x_i\}$ | mean({x}) | median({x }) | mode({x}) | std({x}) | var({x}) or $(\mathrm{std}(\{x\}))^2$ | interquartile({x}): 75 percentile -25 percentile |
| {kx + c} | k*mean({x}) +c | k*median ({x})+c | k*mode({x })+c | \|k\|*std({x }) | $k^2$ * var({x}) | k*interquartile({x}) **if k>0** |

# Summarizing 1D continuous data

- Mean

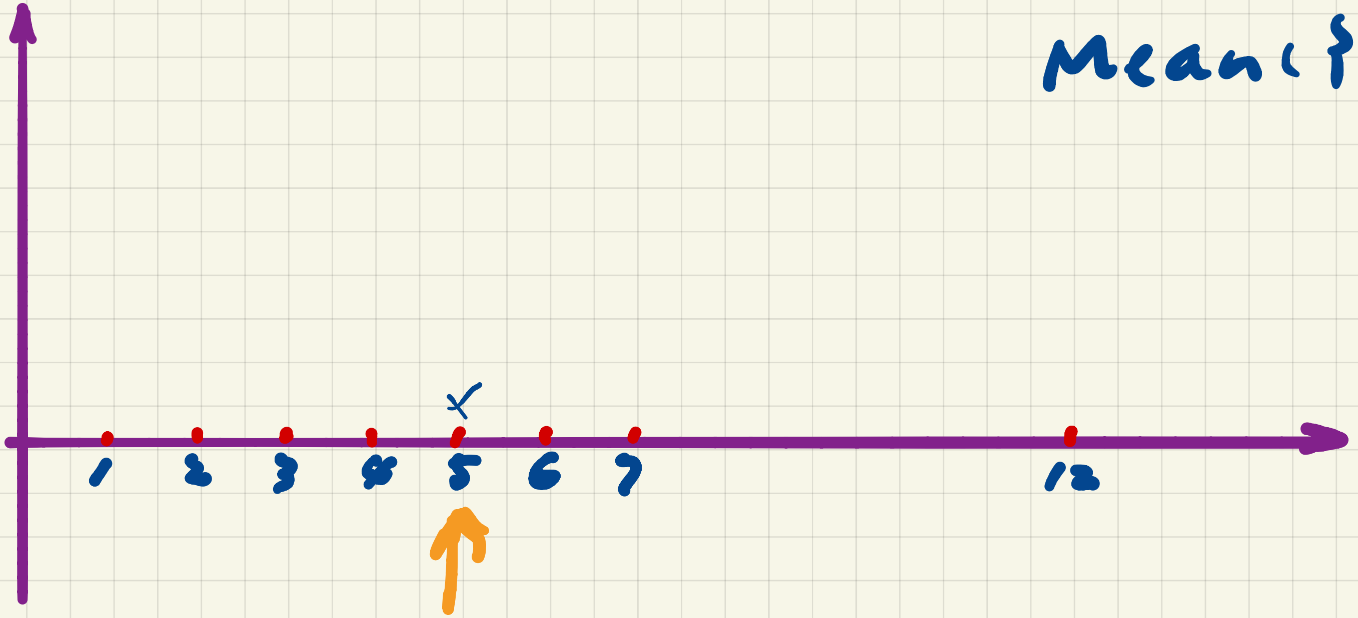
$$mean(\{x_i\}) = \frac{1}{N}\sum_{i=1}^{N} x_i$$

It's the centroid of the data geometrically,
by identifying the data set at that point, you find the center of balance.

$\{x_i\} \quad i \in [1, 8]$

$\{x_i\} = 1, \ 2, \ 3, \ 4, \ 5, \ 6, \ 7, \ 12$

$\text{Mean}(\{x_i\}) = 5$



# Properties of the mean

$\{x_i\}$ $\{y_i\}$

- Scaling data scales the mean

$i \in 1:N$

$mean(\{x_i + y_i\}) = mean(\{x_i\}) + mean(\{y_i\})$

$$mean(\{k \cdot x_i\}) = k \cdot mean(\{x_i\})$$

$mean(\{k \cdot x_i + c\}) = k \cdot mean(\{x_i\}) + c$

- Translating the data translates the mean

$$mean(\{x_i + c\}) = mean(\{x_i\}) + c$$

# Less obvious properties of the mean

- The signed distances from the mean

  sum to 0

$$\sum_{i=1}^{N}(x_i - mean(\{x_i\})) = 0$$

- The mean minimizes the sum of the squared distance from any real value

$$\underset{\mu}{argmin} \sum_{i=1}^{N}(x_i - \mu)^2 = mean(\{x_i\})$$

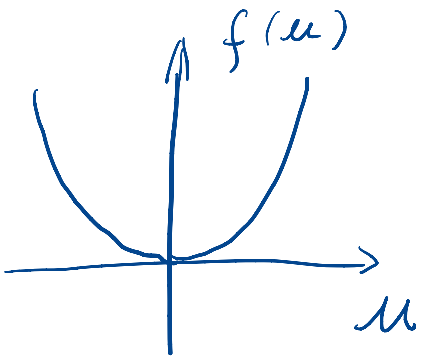
do it at home

# Proof: $\sum_{i=1}^{N}(x_i - \mathrm{mean}(\{x_i\})) = 0$

# Define $\arg\min_{u} f(u)$

Argument $u$ that minimizes the function that follows which is $f(u)$

i.e. if $f(u) = u^2$




$\arg\min_{u} f(u) = ?$

$\hat{u} = \arg\min u^2$

$= 0$

# Proof: $\underset{\mu}{\operatorname{argmin}} \left( \sum_{i=1}^{N} (x_i - \mu)^2 \right) = \operatorname{mean}(\{x_i\})$

Argument $\mu$ that minimizes the function that follows.

$$LHS = \underset{\mu}{\operatorname{argmin}}(f) = \hat{\mu}$$

$$f = \sum_{i=1}^{N} (x_i - \mu)^2$$

$$= \sum_{i=1}^{N} h(\mu) = \sum_{i=1}^{N} g(\mu)^2$$

$\hat{\mu}$ satisfies $\frac{df}{d\mu} = 0$ for a smooth function.

# Proof: $\underset{\mu}{\operatorname{argmin}} \left( \sum_{i=1}^{N} (x_i - \mu)^2 \right) = \operatorname{mean}(\{x_i\})$

Set $\frac{df}{d\mu} = 0$ to find $\hat{\mu}$ $\quad h(\mu) = g^2$

using the chain rule

$$\frac{df}{d\mu} = ? \quad \frac{df}{d\mu} = \frac{d \sum h(\mu)}{d\mu} = \frac{\sum d\, h(\mu)}{d\mu}$$

$$h = g^2 \; ; \quad g = x_i - \mu$$

$$\frac{df}{d\mu} = \sum \frac{d\,h(\mu)}{dg} \cdot \frac{dg}{d\mu} = \sum_i 2g \cdot (-1)$$

$\hookrightarrow 2g$

# Proof: $\underset{\mu}{\operatorname{argmin}} \left( \sum_{i=1}^{N} (x_i - \mu)^2 \right) = \operatorname{mean}(\{x_i\})$

$$\sum_i -2g = 0$$

$$\sum_i -2(x_i - \mu) = 0$$

$$\sum_{i=1}^{N} (x_i - \mu) = 0$$

$$\sum_i^N x_i - \sum_i^N \mu = 0$$

$$\sum_i^N x_i - N \cdot \mu = 0$$

$$\frac{d^2 f(\mu)}{d^2 \mu} ?$$

# Q1:

- What is the answer for

  $mean(\{mean(\{x_i\})\})$ ?

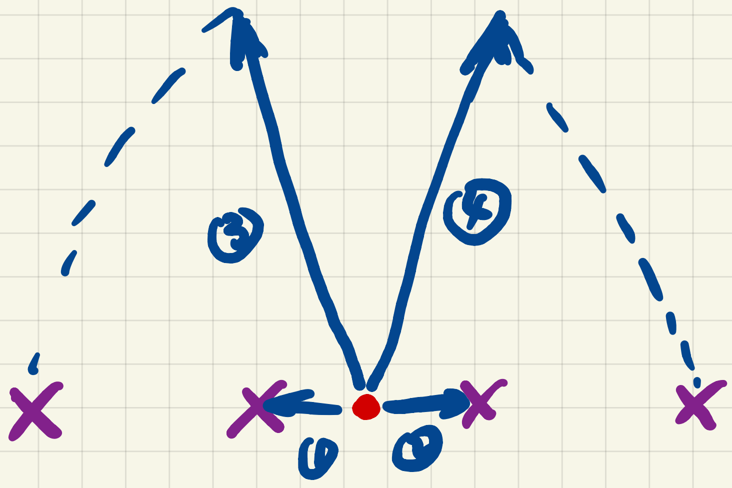
  A. $mean(\{x_i\})$ B. unsure C. 0

# Standard Deviation (σ)

- The standard deviation

$$std(\{x_i\}) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - mean(\{x_i\}))^2}$$

$$= \sqrt{mean(\{(x_i - mean(\{x_i\}))^2\})}$$




How much the data spreads out wrt mean

$$\text{std} = \sqrt{\frac{1}{4}\sum_{i=1}^{4} d_i^{2}}$$

# Q2. Can a standard deviation of a dataset be -1?

A. YES
B. NO

# Properties of the standard deviation

- Scaling data scales the standard deviation

$$std(\{k \cdot x_i\}) = |k| \cdot std(\{x_i\})$$

- Translating the data does **NOT** change the standard deviation

$$std(\{x_i + c\}) = std(\{x_i\})$$

$$std(\{kx + c\}) = |k| \cdot std(\{x\})$$

# Chebyshev's inequality (1st look) using standard deviation:

- At most $\frac{N}{k^2}$ data values are k units of standard deviations away from the mean

k is a number > 1

- Rough justification: Assume mean =0 and

σ is std({x})

assume {−kσ, 0, ...., 0, kσ}

$N - \frac{N}{k^2}$

$\frac{0.5N}{k^2}$    0   # of "0"    $\frac{0.5N}{k^2}$   >

$-k\sigma$      $k\sigma$

$\frac{0.5N}{k^2}(k\sigma)^2 + \frac{0.5N}{k^2}(-k\sigma)^2$

$$std = \sqrt{\frac{1}{N}[(N - \frac{N}{k})0^2 + \frac{N}{k^2}(k\sigma)^2]} = \sigma$$

# Variance ($\sigma^2$)

- Variance = (standard deviation)$^2$

$$var(\{x_i\}) = \frac{1}{N}\sum_{i=1}^{N}(x_i - mean(\{x_i\}))^2$$

- Scaling and translating similar to standard deviation

$$var(\{k \cdot x_i\}) = k^2 \cdot var(\{x_i\})$$

$$var(\{x_i + c\}) = var(\{x_i\})$$

# Q3: Standard deviation

- What is the value of

  $std(\{mean(\{x_i\})\})$ ?

A. 0 B. 1 C. unsure

# Standard Coordinates/normalized data

- The *mean* tells where the data set is and the *standard deviation* tells how spread out it is. If we are interested only in comparing the shape, we could

  define:
  $$\widehat{x_i} = \frac{x_i - mean(\{x_i\})}{std(\{x_i\})}$$

- We say $\boxed{\{\widehat{x_i}\}}$ is in standard coordinates

# Q4: Mean of standard coordinates

- mean( $\{\widehat{x_i}\}$ ) is:

  A. 1 B. 0 C. unsure

$$\widehat{x_i} = \frac{x_i - mean(\{x_i\})}{std(\{x_i\})}$$

# Q5: Standard deviation (σ) of standard coordinates

- Std( $\{\widehat{x_i}\}$ ) is:

  A. 1 B. 0 C. unsure

$$\widehat{x_i} = \frac{x_i - mean(\{x_i\})}{std(\{x_i\})}$$

# Q6: Variance of standard coordinates

- Variance of $\{\widehat{x_i}\}$ is:

  $\boxed{\text{A. 1}}$ B. 0 C. unsure

$$\widehat{x_i} = \frac{x_i - mean(\{x_i\})}{std(\{x_i\})}$$

# Q7: Estimate the range of data in standard coordinates

- Estimate as close as possible, 90% data is within:

  A. [-10, 10]

  B. [-100, 100]

  C. [-1, 1]

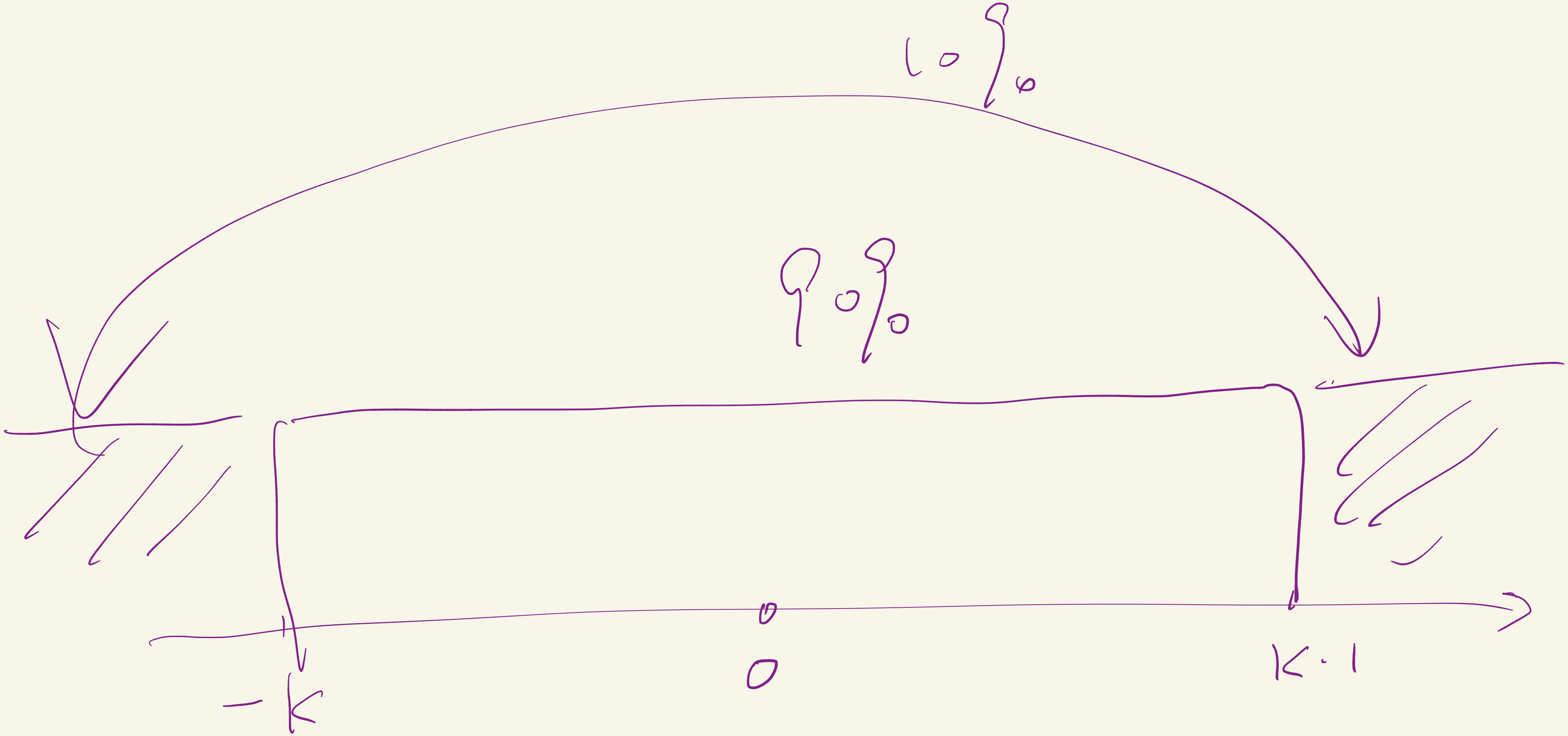
  D. [-4, 4]

  E. others

$$\frac{N}{K^2} = 0.1N$$

$$K \sim \sqrt{10}$$

$$\widehat{x_i} = \frac{x_i - mean(\{x_i\})}{std(\{x_i\})}$$




$\sigma = 1$

$$\frac{\frac{N}{K^2}}{N} = \frac{1}{K^2} \sim 10\% \qquad K \cdot \sigma$$

# Standard Coordinates/normalized data to $\mu$=0, $\sigma$=1, $\sigma^2$=1

- Data in standard coordinates always has

  mean = 0; standard deviation =1;

  variance = 1.

- Such data is unit-less, plots based on this sometimes are more comparable

- We see such normalization very often in statistics

# Median

- We first sort the data set $\{x_i\}$
- Then *if* the number of items N is odd

  median = middle item's value

  *if* the number of items N is even

  median = mean of middle 2 items' values

# Properties of Median

- Scaling data scales the median

$$median(\{k \cdot x_i\}) = k \cdot median(\{x_i\})$$

$$median(\{kx + c\}) = k \cdot median(\{x\}) + c$$

- Translating data translates the median

$$median(\{x_i + c\}) = median(\{x_i\}) + c$$

# Percentile

- $k^{th}$ percentile is the value relative to which k% of the data items have smaller or equal numbers
- Median is roughly the $50^{th}$ percentile

$\{1, 2, 3, 4, 5, 6, 7, 12\}$

75th percentile = ?

$8 \times 0.75$

6

# Assignments

- **HW1** due 9/5 Thurs.
- Reading upto Chapter 2.1
- Next time: the correlation coefficient

# Additional References

- Charles M. Grinstead and J. Laurie Snell "Introduction to Probability"
- Morris H. Degroot and Mark J. Schervish "Probability and Statistics"

# See you next time

*See You!*